



Data Exercise for Selection of Research Assistant under ICSSR Sponsored Research Project

Background

Agriculture remains the principal source of livelihood for most rural households in Bihar, yet productivity continues to lag behind the national average. Despite repeated policy efforts, the diffusion of modern agricultural technologies, ranging from improved seed varieties and micro-irrigation to mobile-based advisory services and mechanisation, has been slow and highly uneven across districts, social groups, and farm sizes.

This study was designed to understand the patterns and correlates of agricultural technology adoption among smallholder farmers in Bihar, and to examine how adoption relates to farm-level outcomes such as crop yield and household income. A primary household survey was administered across selected districts of Bihar, capturing demographic, socio-economic, and agricultural information from 140 farming households.

Assignment

You are expected to analyse the dataset and produce a single policy brief. The four components of the assignment are described below.

1. Clean the Data: Inspect the raw dataset and bring it to an analysis-ready state.
2. Conduct the Statistical and Econometric Analysis: Conduct an appropriate analysis to understand patterns of technology adoption among the surveyed households and to examine its relationship with farm-level outcomes. Preferably, your analysis should make use of:
 - Descriptive summaries that profile the sample
 - Data Visualisations
 - Inferential and econometric methods suitable to the research questions
3. Prepare the Policy Brief: Synthesise your analysis into a policy brief (not more than 1500 words) pertaining to agricultural technology adoption. The brief should:
 - Frame the issue and motivate the analysis in a few short paragraphs

- Present the key empirical findings clearly, supported by figures and/or tables
- Translate the findings into actionable policy recommendations
- Acknowledge important limitations of the data and analysis

4. Referencing and Use of AI Tools: All citations and the reference list must strictly follow APA style. In addition, candidates are required to include a brief disclosure statement at the end of the report acknowledging any use of generative AI tools (if applicable), clearly specifying the nature and extent of such use.

Deliverables

- **Policy Brief (PDF):** Maximum 1,500 words, inclusive of analysis, key findings, and policy recommendations.
- **Supporting Analytical File:** Submit the corresponding working file used for analysis (e.g., Excel workbook, R script, Stata do-file, or any other statistical/analytical file).